Shaunak Bhandarkar

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Education

2024–29 (Expected) **Ph.D. Candidate**, Computational Neuroscience, Princeton University
Advisors: Dr. Jonathan Pillow (Neuroscience) and Dr. Boris Hanin (ORFE)

2020-24 **B.S. with Honors**, Mathematics and Symbolic Systems, Stanford University
Graduated with Distinction (GPA: 4.07)
Visiting student in mathematics, University of Oxford (Spring 2023)

Research Interests: Using theoretical approaches from math and statistics, I aim to characterize fundamental aspects of deep learning: the impact of learning on features and generalization, and the key model dimensions and hyperparameters giving rise to successful learning at large scale. I am also applying deep learning principles to infer learning rules from animal behavior.

Keywords: Neural Networks, Machine Learning Theory, Feature Learning, Learning Rules

Awards & Honors

2026-2029 National Defense Science and Engineering Graduate (NDSEG) Fellowship
2025-26 Princeton Quantitative Neuroscience Training Program Trainee (NIH T32 Grant)
2024 Princeton Centennial Fellowship in the Natural Sciences and Engineering
2024 NSF Graduate Research Fellowship (NSF GRFP) Honorable Mention
2024 Phi Beta Kappa Honor Society
2024 Stanford George Pólya Honors Thesis Prize
2024 Stanford Mathematics Distinguished Service Award

Research Experience

2025–Present **Princeton Neuroscience Institute**, Princeton, NJ
Graduate Researcher. Advisors: Dr. Jonathan Pillow and Dr. Boris Hanin

- Studying how gradient descent shapes task-relevant features in neural networks. Proved that one step of gradient descent in a deep linear network adds a rank-one, target-aligned component to each layer's feature kernel, and derived exact conditions on a low-rank (spiked) data model under which these features improve generalization. With B. Hanin; preprint posted on arXiv, 2026.
- Developed a statistical method, implemented in JAX, for inferring which learning rules in shallow and deep networks can explain trial-by-trial animal behavior (stimuli, choices, rewards). Across 58 mice (~10,000 trials each), adding depth and nonlinearity yielded best fits. With J. Pillow; paper to appear at NeurIPS 2026.

2022–24 **Stanford University**, Stanford, CA
Undergraduate Researcher. Advisor: Dr. James L. McClelland
• Developed biologically plausible neural network architectures for achieving the same functional outcomes as contemporary memory- and attention-based AI systems (e.g., Transformers), suggesting how the human mind may support similar complex computations as its AI counterparts. Preprint on arXiv, 2025.

2023–24 **Stanford University**, Stanford, CA
Undergraduate Honors Thesis [thesis]. Advisor: Dr. Scott W. Linderman
• Analyzed a class of "neuromodulated" recurrent networks (NM-RNNs), in which a small network dynamically scales the low-rank weights of a larger network; formalized the connection between NM-RNNs and LSTMs, and demonstrated how an NM-RNN splits a task between its two sub-networks. Paper at NeurIPS 2024.

Summer 2023 **Flatiron Institute – Center for Computational Neuroscience**, New York, NY
Undergraduate Research Assistant. Supervisor: Dr. SueYeon Chung
• Studied how one gradient step changes representational geometry depending on the size of the step.

Publications and Abstracts

Peer-reviewed Conference Proceedings

1. **Bhandarkar, S.** & Pillow, J. *Inferring learning rules in deep neural network architectures from animal learning data.* The Fortieth Annual Conference on Neural Information Processing Systems 2026. [Accepted / Forthcoming].
2. Costacurta, J. C., **Bhandarkar, S.**, Zoltowski, D. M. & Linderman, S. W. *Structured flexibility in recurrent neural networks via neuromodulation.* The Thirty-eighth Annual Conference on Neural Information Processing Systems 2024. [paper].

Preprints

1. **Bhandarkar, S.** & Hanin, B. *Investigating the Early Signatures of Feature Learning in Deep Linear Networks.* arXiv:2610.37834, 2026. [paper].
2. **Bhandarkar, S.** & McClelland, J. *Neural Computation Without Slots: Steps Towards Biologically Plausible Memory and Attention in Natural and Artificial Intelligence.* arXiv:2511.04593, 2025. [paper].

Conference Abstracts and Workshop Papers

1. Costacurta, J., **Bhandarkar, S.**, Zoltowski, D. & Linderman, S. *Neuromodulated Recurrent Neural Networks.* Computational and Systems Neuroscience (COSYNE). 2024. [poster].
2. **Bhandarkar, S.** & McClelland, J. *Sequential Learning and Retrieval in a Sparse Distributed Memory: The K-winner Modern Hopfield Network.* NeurIPS Workshop on Associative Memory & Hopfield Networks. 2023. [manuscript + oral presentation].

Presentations

Talks

1. **Bhandarkar, S.** *"Sequential Learning and Retrieval in a Sparse Distributed Memory: The K-winner Modern Hopfield Network."* Bay Area Memory Meeting (BAMM!) (Stanford, CA). 2023.

Posters

1. **Bhandarkar, S.** & Hanin, B. *Investigating the Early Signatures of Feature Learning in Deep Linear Networks.* Harvard Kempner Institute Workshop on Learning Dynamics in Natural and Artificial Intelligence. 2026. [poster].
2. **Bhandarkar, S.** & Cohen, J. *Attention-based Memory Supports Learning of Semantically Rich Representations.* Princeton Machine Learning Theory Summer School. 2025. [poster].
3. **Bhandarkar, S.**, Chou, C.-N. & Chung, S. *The Geometry of Learning: How One Gradient Step Alters Latent Manifold Structure.* Flatiron Institute Summer Research Internship. 2023. [poster].

Teaching

Princeton University

2026 Teaching Assistant, Introduction to Cognitive Neuroscience (NEU 202)
- Introducing undergraduates to topics in cognitive neuroscience; commitments include holding precepts for discussing papers and grading.

2025 Teaching Assistant, Mathematical Tools for Neuroscience (NEU 314)
- Introducing undergraduate neuroscience majors to quantitative neuroscience topics (programming, linear algebra, dynamical systems, probability); commitments include holding review/help sessions, creating assignments and exams, and grading.

Skills

Programming Languages: Python (JAX, PyTorch), C++, MATLAB, Java

Tools/Frameworks: Git, Slurm computing clusters, LaTeX

Training Experiences: University of Michigan Random Matrix Theory Summer School (2026), Princeton Machine Learning Theory Summer School (2025)

Community Involvement

2016-24 **Head Instructor at Special Needs Aquatic Program (SNAP)**, Palo Alto, CA
- Helped kids with special needs exercise their muscles and have fun in the Palo Alto YMCA pool. Led group swim activities and recruited new Stanford volunteers.

2022-24 **Service for Stanford Math Department**, Stanford, CA
- Served as a peer advisor (2022–24), undergraduate tutor (2023–24), and a Stanford University Math Camp (SUMaC) teaching assistant (Summer 2022).